HANDWRITTEN CHARACTER RECOGNITION SYSTEM

Machine Learning Project Phase II – Proposal Document

Submitted By

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Project Title

CNN-Based Handwritten Character Recognition System

1. Background and Motivation

Handwritten character recognition is an important area in computer vision and machine learning, particularly in document analysis and Optical Character Recognition (OCR). It involves automatically identifying handwritten characters such as letters and digits from image-based inputs.

The motivation behind this project is to explore the application of Convolutional Neural Networks (CNNs) for handwritten character classification. Handwritten text varies significantly due to differences in writing styles, stroke thickness, orientation, and character formation. Traditional rule-based systems struggle to handle such variations effectively, making deep learning-based approaches more suitable.

This project provides practical experience in image classification, convolutional neural networks, preprocessing techniques, and model evaluation while addressing a real-world problem with applications in digitization and automated text recognition systems.

2. Problem Statement

The objective of this project is to design and implement an automated handwritten character recognition system capable of correctly identifying handwritten characters from grayscale images.

The system aims to classify handwritten digits and letters despite variations in writing styles and image characteristics. A CNN-based deep learning model will be developed and trained to recognize different character classes with high accuracy.

The project addresses the challenge of feature extraction and classification from handwritten image data while minimizing prediction errors and improving classification performance.

3. Selected Dataset and Dataset Link

   Dataset Name:

   EMNIST Balanced Dataset

   Dataset Source:

   https://www.kaggle.com/datasets/crawford/emnist

   Reason for Selection:

   The EMNIST dataset was selected because it extends the traditional MNIST handwritten digit dataset by including both letters and digits. It provides a balanced set of handwritten characters and is widely used as a benchmark dataset for handwritten recognition tasks.

4. Dataset Description

   The project uses the EMNIST Balanced dataset, which contains handwritten characters collected from multiple writers.

   Dataset Statistics

   Total Images: 131,599

   Training Images: 112,800

   Testing Images: 18,800

   Number of Classes: 47

   Image Resolution: 28 × 28 pixels

   Image Type: Grayscale

   Class Labels:

   Digits (0–9) and handwritten letters

   Class Distribution:

   The EMNIST Balanced dataset contains relatively balanced samples across all classes to reduce classification bias during training.

5. Research Questions

The following research questions are addressed in this project:

Can a CNN-based model effectively recognize handwritten characters with high accuracy?

How well can the model generalize across different handwriting styles?

What level of classification performance can be achieved using the EMNIST Balanced dataset?

How effective are convolutional neural networks for feature extraction in handwritten image classification tasks?

6. Expected Results

The expected outcome of this project is a CNN-based handwritten character recognition model capable of achieving strong classification accuracy on unseen handwritten character images.

The system is expected to:

Correctly classify handwritten characters from grayscale images

Achieve good classification performance on test data

Generate evaluation metrics such as accuracy, loss, confusion matrix, and classification reports

Provide visualizations including training curves and prediction performance analysis

7. Proposed Methodology

The methodology for this project follows a structured deep learning workflow.

Step 1: Dataset Collection

The EMNIST Balanced dataset will be collected and loaded into the development environment.

Step 2: Data Preprocessing

The dataset will undergo preprocessing including:

Image normalization

Image reshaping for CNN compatibility

Correction of EMNIST image orientation

Train-test split preparation

Step 3: CNN Model Development

A Convolutional Neural Network (CNN) architecture will be developed for handwritten character classification.

The proposed architecture includes:

Convolutional Layer (32 filters, 3×3 kernel, ReLU activation)

Max Pooling Layer (2×2)

Convolutional Layer (64 filters, 3×3 kernel, ReLU activation)

Max Pooling Layer (2×2)

Flatten Layer

Dense Layer (128 neurons, ReLU activation)

Dropout Layer (0.3)

Output Layer (47 neurons, Softmax activation)

Step 4: Model Training

The CNN model will be trained using:

Optimizer: Adam

Loss Function: Sparse Categorical Crossentropy

Epochs: 8

Batch Size: 32

Validation Split: 20%

Step 5: Model Evaluation

The trained model will be evaluated using multiple performance metrics and visualization techniques.

8. Transfer Learning Models

To compare CNN performance, transfer learning approaches may be explored using pre-trained image classification architectures such as MobileNetV2 or ResNet50.

The performance of the custom CNN model will be compared with at least one transfer learning model to analyze differences in accuracy and computational efficiency.

9. Evaluation Metrics

The following evaluation metrics will be used to assess model performance:

Accuracy

Loss

Confusion Matrix

Classification Report

Precision

Recall

F1-score

Training and Validation Curves

10. Expected Figures and Tables

The project is expected to include the following outputs:

Sample handwritten character visualizations

Accuracy curve

Loss curve

Confusion matrix

Classification report table

Model performance comparison table

Training and validation performance graphs

11. Conclusion

This proposal presents a CNN-based handwritten character recognition system using the EMNIST Balanced dataset. By leveraging convolutional neural networks and image preprocessing techniques, the project aims to build an effective handwritten character classification system capable of handling different writing styles and character variations.

The project will provide practical insights into image classification, deep learning workflows, and performance evaluation for handwritten recognition tasks.